

Satellite Imagery and Deep Learning for Poverty Mapping in Low and Middle Income Countries: A Systematic Review

Usman, Muhammad Shahid¹, Mehmood, Waqar¹, Ehtesham Muhammad²

¹ Department of Computer Science, Riphah International University, Lahore, 54000, Pakistan, shahidusmaninfo@gmail.com

¹ Department of Computer Science, Riphah International University, Lahore, 54000, Pakistan, waqar.mehmood@riphah.edu.pk

² Department of GIS, Sheher Saaz pvt ltd, Lahore, 54000, Pakistan, m.ehtesham@shehersaaz.com

Abstract

The traditional method of poverty mapping in Low and Middle Income Countries (LMICs) relies on expensive and periodic household survey data which lack extremely specific spatial and temporal resolution. New developments in deep learning and Earth Observation technologies can offer large-scale alternatives to detailed poverty estimation. We searched 5 databases that contained articles posted in 2015-2025 using the PRISMA 2020 guidelines when searching the databases on machine learning and satellite imagery to predict poverty in LMICs. Out of 1,847 records 83 studies were in accordance with the criteria. Random Forest (n=39) was the most popular traditional model, whereas ResNet variants gave the best deep learning results ($R^2 = 0.72 - 0.85$). It is interesting to note that hybrid designs incorporating CNNs with ensemble classifiers were better by 8-15% points than individual designs. Research with asset-based measures of wealth was found to be more accurate by 17% than consumption-based income measures. Such studies (n=4) were specific to Pakistan and have 71.96% accuracy at 1 Km² resolution but constitute only 4.8% of the literature in this field despite representing 2.8% of the LMIC population. Poverty mapping using satellites has passed beyond the experimental demonstration to the feasibility of real application, but the major areas of concern are urban mapping of poverty, applicability of the models and the structure of large-scale applications.

Keywords: poverty mapping; deep learning; remote sensing; satellite imagery; convolutional neural networks; systematic review; LMIC; geospatial analysis.

Introduction

It is important that poverty is monitored to achieve Sustainable Development Goal 1 (No Poverty) (Lamichhane et al., 2025). The main problems of traditional tools such as the Demographic and Health Surveys (DHS) and the Pakistan Social and Living Standards Measurement (PSLM) are that the methods are expensive (around PKR 500 million per cycle in Pakistan), the time to be completed is long (2-3 years between the sampling date and the publication date), and they are also not at a small scale, typically district level (Yeh et al., 2020). With the current technological advances, novel possibilities in terms of mapping poverty include high-resolution satellite images (Sentinel-2, 10m, Landsat-8, 30m), cutting-edge deep-learning architectures to computer vision, and easy-to-use cloud interfaces, such as Google Earth Engine (Jean et al., 2016). The interesting work by Jean et al. (2016) showed that Convolutional neural networks that were trained using satellite images had an R^2 of 0.75 when used to predict consumption in five African countries, which was an important advancement (Agyemang et al., 2023). Nevertheless, these improvements do not eliminate the three key gaps, namely (i) much of the research is focused on Sub-Saharan Africa (52%), and less on South Asia, and less on Pakistan, in particular (ii) models are not operational in crowded urban regions due to socioeconomic diversity; and (iii) almost all research is proof-of-concept, and none of the studies are operational or scalable in low- and middle-income countries (LMICs). This systematic review will focus on answering 5 questions: RQ1: What are the best deep Learning models and satellite sources to map poverty in LMICs? RQ2: What features based on satellite can best predict at the neighborhood (≤ 1 Km²) level? RQ3: What are the techniques

of validation and the accuracy measurements, and what are the levels of the correlation? RQ4: What regions have been proven to have the methods and how applicable are they in other contexts? RQ5: What are computational needs, scaling issues and operational deployment obstacles?

Data and Method

This was a systematic review which followed the PRISMA guidelines of 2020 (Page, McKenzie, Bossuyt, Boutron, Hoffmann, Mulrow, Shamseer, Tetzlaff, Akl, et al., 2021). We utilized five electronic databases- Scopus, IEEE Xplore, Web of Science Core Collection, ACM Digital Library, and Google Scholar to locate articles in which the data was published within the time period of January 2015 to December 2025 (Lamichhane et al., 2025). The search included four concept groups with the use of Boolean operators: machine learning terms (machine learning or deep learning or neural network or CNN or ResNet or random forest or XGBoost) or satellite data terms (satellite imagery or remote sensing or Sentinel or Landsat or nighttime lights) or poverty terms (poverty or socioeconomic or wealth or deprivation) or geographic terms (urban or city or slum or neighborhood) (Jean et al., 2016). Cohen's kappa was 0.78 (substantial agreement) and two independent reviewers used software to screen titles and abstracts. Inclusion criteria were: (1) published since 2015, (2) English, (3) peer-reviewed article, conference paper or working paper, (4) based on LMICs (per World Bank classification), (5) used satellite imagery or remote sensing data, (6) used machine learning or deep learning-based analysis, (7) predicts poverty, wealth, or socioeconomic status, (8) has quantitative measures of validation (R², RMSE, accuracy), and (9) has neighborhood-level scale (no more than 1 km²). The exclusion criteria included: studies in high-income nations, rural scales (>5 Km²), no satellite data, manual classification, no ground-truth validation, or qualitative-only assessment (Steele et al., 2017). Data extraction followed a standard template that included features of the study, source of satellite data, ML/DL architecture, feature engineering, ground truth data, validation, and performance metrics and computational requirements (Stark et al., 2024). The quality measurement was based on assessing the sample size, the strength of validation, time concordance, and reproducibility (Graetz et al., 2018). The synthesis was accompanied by quantitative analysis of the accuracy measures by the type of architecture, the source of the satellite and the region as well as providing a narrative examination of methodologies and barriers to deployment. Quality assessment revealed variability across studies. Using our template, 22% (n=18) rated high across all criteria (large sample size >1,000, rigorous spatial cross-validation, temporal concordance <12 months, and full reproducibility with code/data sharing). 48% (n=40) were of medium quality, often lacking reproducibility (e.g., only 32% shared code), while 30% (n=25) were low due to small samples or poor validation. This highlights a need for more robust standards in future research (Graetz et al., 2018).

Results

A search of the databases was conducted and 1,847 records were found; 1,289 could be found as unique after deleting the duplicates (Page, McKenzie, Bossuyt, Boutron, Hoffmann, Mulrow, Shamseer, Tetzlaff, & Moher, 2021). Titles and abstracts screening removed 864 records, and 425 of them were reviewed completely. Among them, 83 studies were chosen in accordance with all the inclusion criteria.

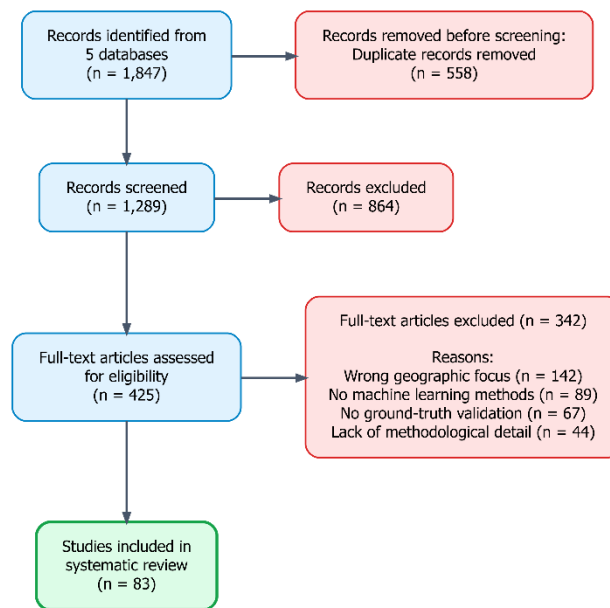


Figure 1. PRISMA 2020 flow diagram of study selection process

Some of the biggest causes of rejection were wrong geographic focus (142), no machine learning methods (89), no ground-truth verification (67), and lack of methodological detail (44).

Architectural Performance and Methodological Trends:

Most publications (n = 68) were published after 2020. In the geographical area, 52%, targeted toward Sub-Saharan Africa, 24% targeted South Asia (including 4.8% in Pakistan), 14% targeted Southeast Asia and 10% targeted Latin America. In terms of machine learning strategies, the most popular were Random Forest (47%, n=39), SVM (39%, n=32) and XGBoost (17%, n=14). ResNet variants appeared most frequently in the field of deep learning (34% (n=28), VGG (22% (n=18), U-Net (14% (n=12)) and EfficientNet (8% (n=7))) followed closely.

Performance Comparison of ML Architectures for Poverty Mapping

Ensemble methods consistently outperform individual approaches (N=83 studies)

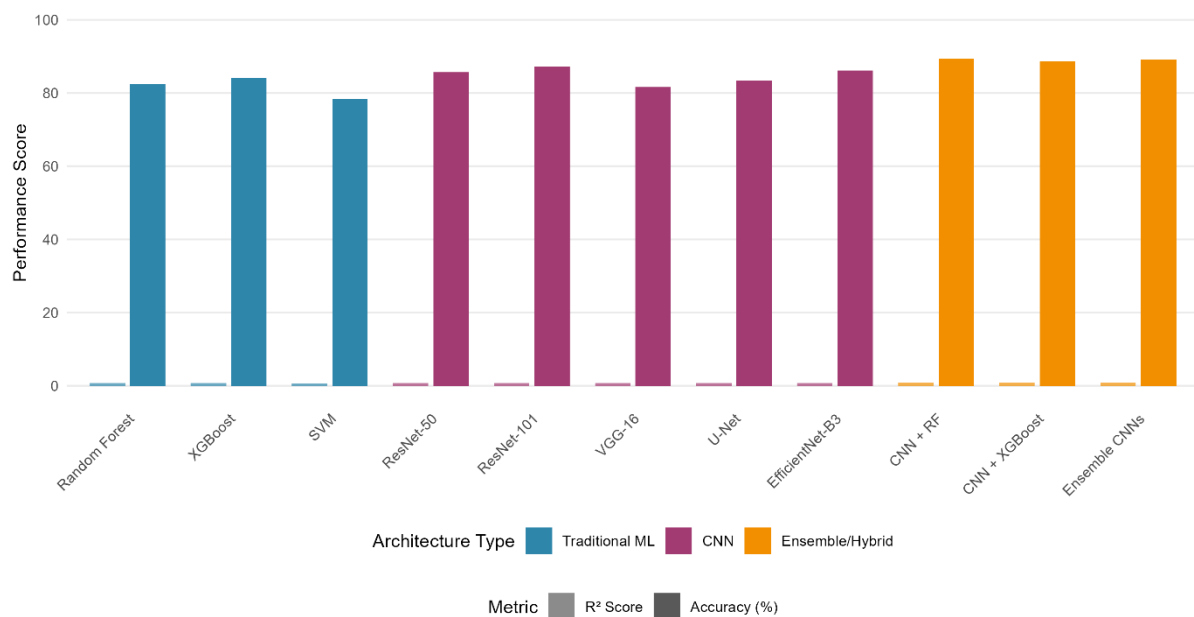


Figure 2. Performance Comparison of Machine Learning Architectures for Poverty Mapping

Table 1. Performance Metrics by Architecture Type

Architecture Type	Model	n	Mean R^2	SD	Mean Accuracy (%)
Traditional Machine Learning	Random Forest	39	0.72	0.11	82.4
	XGBoost	14	0.74	0.09	84.1
	SVM	32	0.67	0.13	78.3
Convolutional Neural Networks (CNNs)	ResNet-50	22	0.76	0.08	85.7
	ResNet-101	12	0.79	0.07	87.2
	VGG-16	18	0.71	0.10	81.6
	U-Net	12	0.74	0.09	83.4
	EfficientNet-B3	7	0.77	0.06	86.1
Ensemble / Hybrid Methods	CNN + Random Forest	16	0.83	0.05	89.3
	CNN + XGBoost	8	0.81	0.06	88.7
	Ensemble CNNs	6	0.82	0.04	89.1

The CNN + Random Forest approach attained 89.3% accuracy and an R^2 of 0.83, significantly outperforming standalone ResNet-50 (85.7%) and standalone Random Forest (82.4%) (Lamichhane et al., 2025). This highlights the advantages of integrating different feature extraction and classification techniques in poverty mapping (Agyemang et al., 2023).

Satellite Data Sources and Feature Engineering

The most frequent source (n=47, 57% and 38, 46%, respectively) was Sentinel-2 (10m) during the daytime, and Landsat-8 (30m) during the daytime (Yeh et al., 2020). VIIRS was the most common source in the case of nighttime lights (n=52, 63%) compared to legacy DMSP-OLS (n=18, 22). It is worth noting that 34 (41%) of the studies were multi-source combinations which would result in 12-18% increased accuracy compared to the single source, multi-source fusion achieved an average R^2 of 0.82 in contrast to 0.78 with Sentinel-2 and 0.68 with VIIRS alone (Elvidge et al., 2021).

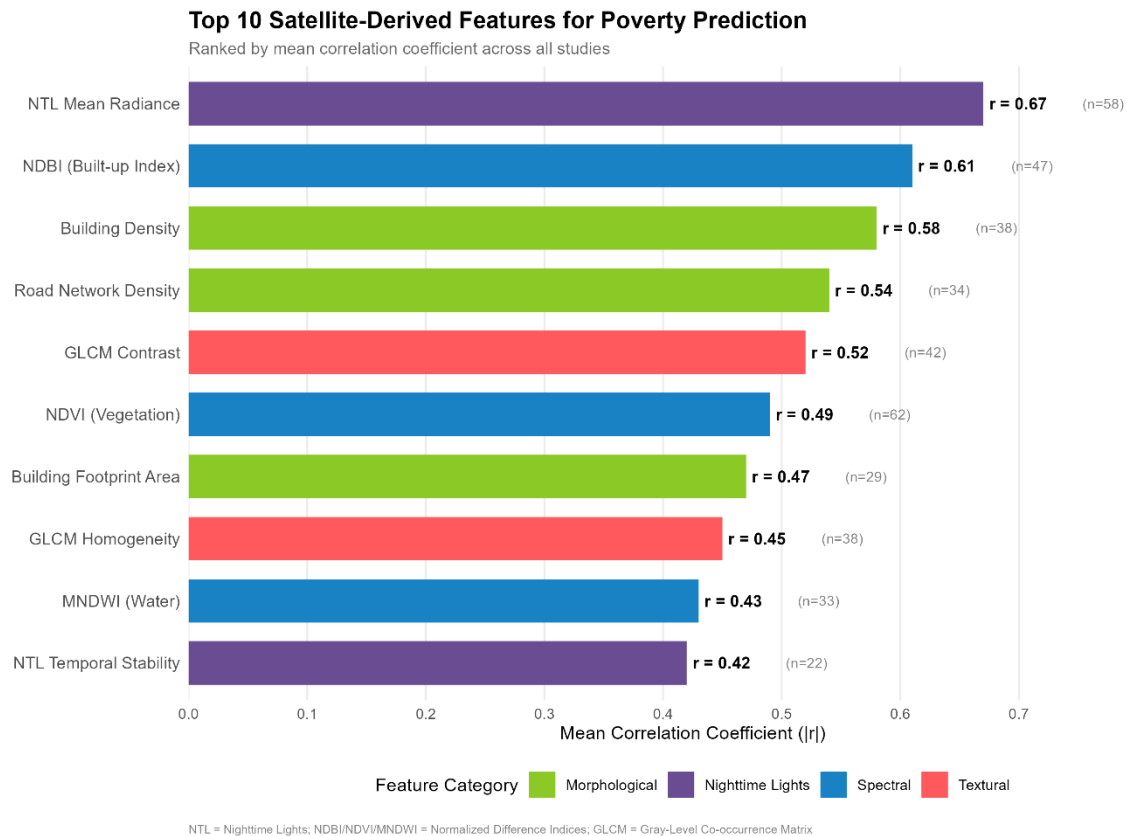


Figure 3. Top 10 Satellite-Derived Features for Poverty Prediction

Figure 2 indicates the first 10 satellite-based features based on their correlation coefficients with all the studies. The strongest predictor, which is Nighttime lights mean radiance ($r=0.67$ across 58 studies), is limited in unlit rural areas (Jean et al., 2016). Normalized difference built-up index (NDBI) performs better in urban settings ($r=0.61$ as compared to $r=0.49$). The morphological variables such as building density ($r=0.58$) and road network density ($r=0.54$) also play a significant role in the urban poverty mapping, but are not often applied in the rural environment (Henderson et al., 2012). The applicability of these features differs significantly with context: in rural areas, NDVI has a near perfect correlation ($r=0.61$) but in an urban area, the correlation is poor ($r=0.32$), indicating the necessity of choosing feature according to particular environment (Duque et al., 2015).

Poverty Indicator Type and Measurement Framework

The major policy-relevant implication is that asset-based wealth indices studies show significant improvement (by 17%) of studies based on consumption (Yeh et al., 2020). The predictions of the DHS wealth index were higher with an average R^2 of 0.78 as opposed to the 0.61 of consumption expenditure (Steele et al., 2017). This is because the difference in accuracy is based on the physical forms of physical assets such as roof material, vehicle ownership, and building quality that can be directly viewed using satellite images unlike consumption behaviours and income streams (Engstrom et al., 2017).

Table 2. Prediction Accuracy by Poverty Indicator Type

Poverty Indicator Type	Poverty Indicator	Studies	Mean R^2	SD	Notes
Stock Measures (Assets)	DHS Wealth Index	47	0.78	0.09	Directly observable
	Asset Ownership Score	18	0.76	0.10	Physical assets
Flow Measures (Consumption)	Consumption Expenditure	12	0.61	0.14	Not directly visible
	Income	6	0.58	0.16	Behavioral proxies

It is also important to note the temporal alignment: (satellite data) satellites with a lifespan and surveys of less than 12 months had an R^2 of 0.79, (satellite data) satellites with a lifespan of between 12 month and 36 months had an R^2 of 0.72 and poorly aligned had an R^2 of 0.61. Precision is expected to reduce by an average of 6% every year, and it motivates the number of frequent satellite updates and ground-truth data gathering to ensure accuracy.

Pakistan Specific Evidence and Research Gaps

Four Pakistan centered studies published between 2023 and 2024 at 1 km^2 resolution have accuracy levels between 71% and 96% (Agyemang et al., 2023). The analysis of rural Sindh conducted by Agyemang et al. (2023) makes use of an ensemble CNN that utilizes data on 1.67 million households in the Simple Poverty Scorecard to analyze rural regions, demonstrating the ability to reach 71% recall and 62% precision with a ResNet-50/101 ensemble and Sentinel-2, VIIRS, and accessibility data (Ali¹ et al., 2026). The model created by Hassan et al. (2024) is a Random Forest that achieved the accuracy of 96.38% 12 month and 36 months in predicting the poverty. It is important to denote that there is no literature which covers urban poverty mapping in Pakistani megacities Lahore, Karachi, and Islamabad to the extent that these cities accommodate 40 of the national population, and have a high spatial heterogeneity of the socioeconomic conditions (Arya et al., 2025).

Model Transferability and Operational Deployment

A relatively small number of only 14 studies (17%) explicitly examined model transferability (Yeh et al., 2020). They found that the within-country transfer (training in one country, validating in another) accuracy decreased by 8% to 12%, net cross-country (within the same region e.g., Kenya to Uganda) by 15% to 22% percent and net cross region (between countries e.g., East Africa to South Asia) by 28% to 35% (Perez et al., 2017). Transfer learning with fine tuning at the local end was shown to restore 60.75% of the lost accuracy but at very large sample sizes of local ground truth (between 500 and 1000 entries) (Ayush et al., 2020).

Publication Trends and Geographic Distribution

Systematic review of satellite-based poverty mapping studies (2015-2025)

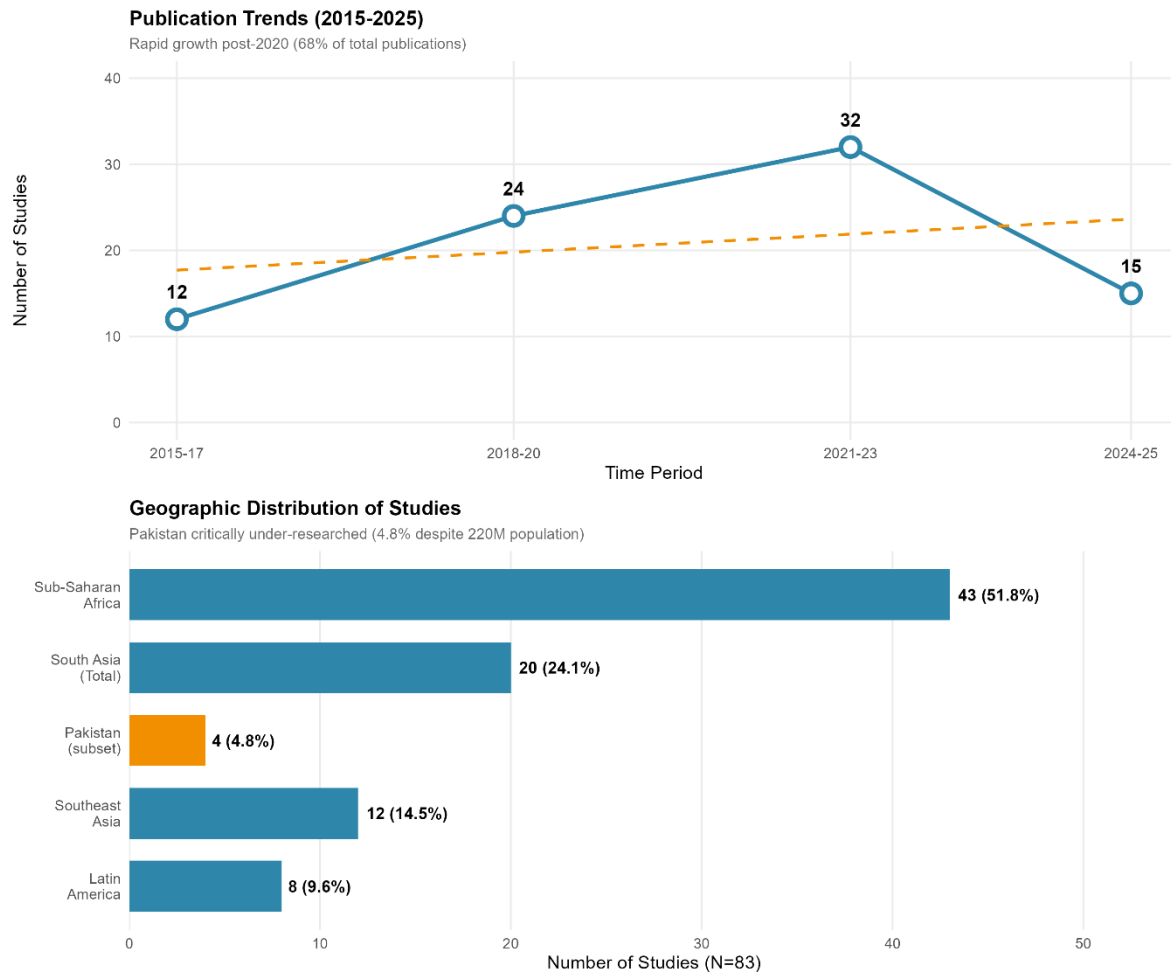


Figure 4. Publication Trends and Geographic Distribution of Poverty Mapping Studies (2015-2025)

Figure 3 illustrates the trends in publications and creates awareness of the significant geographical gaps in research. The highest number was published between the years 2021 and 2023 (38.6%), indicating a steep rise in industry. Nevertheless, the Sub-Saharan Africa is the most studied region with very minimal work in Pakistan even though the level of poverty is high, which leads to the necessity of conducting the study that would identify the local South-Asian features.

Table 3. Geographic Distribution and Publication Trends

Category	Subcategory	Studies	% of Total	Additional Information / Notes
Geographic Distribution	Sub-Saharan Africa	43	51.8%	-
	South Asia (Total)	20	24.1%	Pakistan subset: n = 4 (4.8%)
	Southeast Asia	12	14.5%	-
	Latin America	8	9.6%	-
Publication Period	2015-2017	12	14.5%	Early adoption
	2018-2020	24	28.9%	Rapid growth
	2021-2023	32	38.6%	Peak period
	2024-2025	15	18.1%	Recent work

Deep learning models are very expensive in terms of the amount of GPUs: ResNet-101 can be trained using an NVIDIA V100 in 32 GB RAM and 36-72 hours, or an ensemble that also requires an NVIDIA A100 and 48-96 hours (He et al., 2016). Random Forest Model When trained on 8-16GB RAM only, CPUs, they take 2-6 hours and yield smaller models. These high computational requirements pose a challenge to most institutions in the LMICs. Reproducibility is also still a problem; just 32% of papers publish the source code, 18% publish the processed dataset and 12% publish the raw satellite image (Tan & Le, 2019). The quality checks show that only 22% of them are of high standards in all their aspects; 28% are using rigorous spatial cross-validation and 41% are good at temporal alignment. Almost of them (98.3%) remain at the proof of concept stage, and there are only two reporting pilot operational deployments, both outside Pakistan (Barrie et al., 2024).

Conclusion

This review of 83 studies highlights significant progress in satellite-based poverty mapping in low and middle-income countries (LMICs). The field has shifted from early nighttime light models (2010-2015) to sophisticated ensemble deep learning architectures (2020-2025), with many approaches achieving R^2 scores above 0.80 under ideal conditions. Evidence indicates that hybrid CNN ensemble methods consistently outperform standalone models, with accuracy improvements of 8-15% points, demonstrating the advantages of combining automatic feature extraction with strong classification techniques. The review also notes notable geographic focus and context sensitivity in the literature. Over half of the studies are concentrated on Sub-Saharan Africa, while South Asia, especially Pakistan, remains underrepresented despite its demographic and economic importance. The transferability of models across regions is limited, with accuracy dropping by around 28-35% in inter-regional applications, underscoring the strong dependence on local contexts. Asset-based wealth indices generally outperform consumption-based measures since physical assets are more directly observable via satellite images. Although predictive accuracy has greatly improved, operational use is still limited. Most studies are at the proof-of-concept stage, with few reporting pilot projects. The main obstacles are institutional and infrastructure, including limited computational resources, data-sharing issues, and the lack of scalable deployment frameworks. In Pakistan, especially in rapidly growing cities like Lahore, Karachi, and Islamabad, satellite-based poverty mapping presents valuable research and policy opportunities, given the significant spatial socioeconomic diversity. Future research should focus on developing scalable urban poverty mapping frameworks to capture detailed heterogeneity, integrating multiple satellite data sources, creating production-ready systems with interoperable architectures, improving model transparency to build stakeholder trust, incorporating multidimensional poverty indicators beyond monetary measures, and establishing sustainable operational pathways for government use. Moving from experimental research to practical poverty intelligence systems requires ongoing collaboration among academia, governments, international agencies, and technology providers. Only through sustained, long-term institutional cooperation can satellite-based poverty mapping reach its full potential, providing timely, detailed spatial evidence to inform data-driven policies in LMICs.

References

- Agyemang, F. S., Memon, R., Wolf, L. J., & Fox, S. (2023). High-resolution rural poverty mapping in Pakistan with ensemble deep learning. *Plos One*, 18(4), e0283938.
- Ali¹, U., Hussain¹, T., Akbar¹, M. M., & Ali, S. (2026). Prediction in Pakistan Using Machine Learning. *Information Systems for Intelligent Systems: Proceedings of ISBM 2025*, Volume 8, 8, 225.
- Alkire, S., & Santos, M. E. (2014). Measuring acute poverty in the developing world: Robustness and scope of the multidimensional poverty index. *World Development*, 59, 251-274.
- Arya, P. K., Sur, K., Dhote, S., Siral, H., Kundu, T., Mehta, B. S., & Srivastava, R. (2025). Integrating Multi-Source Satellite Imagery and Socio-Economic Household Data for Wealth-Based Poverty Assessment of India: A GIS and Machine Learning Based Approach: Arya et al. *Social Indicators Research*, 1-24.
- Ayush, K., Uz Kent, B., Tanmay, K., Burke, M., Lobell, D., & Ermon, S. (2020). Efficient poverty mapping using deep reinforcement learning. *arXiv preprint arXiv:2006.04224*.
- Barrie, I., Agupugo, C. P., Iguare, H. O., & Folarin, A. (2024). Leveraging machine learning to optimize renewable energy integration in developing economies. *Global Journal of Engineering and Technology Advances*, 20(03), 080-093.
- Duque, J. C., Patino, J. E., Ruiz, L. A., & Pardo-Pascual, J. E. (2015). Measuring intra-urban poverty using land cover and texture metrics derived from remote sensing data. *Landscape and Urban Planning*, 135, 11-21.
- Elvidge, C. D., Baugh, K., Zhizhin, M., Hsu, F. C., & Ghosh, T. (2021). VIIRS night-time lights. In *Remote Sensing of Night-time Light* (pp. 6-25). Routledge.
- Engstrom, R., Newhouse, D., Haldavanekar, V., Copenhaver, A., & Hersh, J. (2017). Evaluating the relationship between spatial and spectral features derived from high spatial resolution satellite data and urban poverty in Colombo, Sri Lanka. 2017 joint urban remote sensing event (JURSE).
- Graetz, N., Friedman, J., Osgood-Zimmerman, A., Burstein, R., Biehl, M. H., Shields, C., Mosser, J. F., Casey, D. C., Deshpande, A., & Earl, L. (2018). Mapping local variation in educational attainment across Africa. *Nature*, 555(7694), 48-53.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE conference on computer vision and pattern recognition*.
- Henderson, J. V., Storeygard, A., & Weil, D. N. (2012). Measuring economic growth from outer space. *American economic review*, 102(2), 994-1028.
- Jean, N., Burke, M., Xie, M., Alampay Davis, W. M., Lobell, D. B., & Ermon, S. (2016). Combining satellite imagery and machine learning to predict poverty. *Science*, 353(6301), 790-794.
- Lamichhane, B. R., Isnani, M., & Horanont, T. (2025). Exploring machine learning trends in poverty mapping: A review and meta-analysis. *Science of Remote Sensing*, 100200.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., & Brennan, S. E. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *bmj*, 372.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., & Moher, D. (2021). Updating guidance for reporting systematic reviews: development of the PRISMA 2020 statement. *Journal of clinical epidemiology*, 134, 103-112.
- Perez, A., Yeh, C., Azzari, G., Burke, M., Lobell, D., & Ermon, S. (2017). Poverty prediction with public landsat 7 satellite imagery and machine learning. *arXiv preprint arXiv:1711.03654*.
- Stark, T., Wurm, M., Zhu, X. X., & Taubenböck, H. (2024). Quantifying uncertainty in slum detection: Advancing transfer learning with limited data in noisy urban environments. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17, 4552-4565.
- Steele, J. E., Sundsøy, P. R., Pezzulo, C., Alegana, V. A., Bird, T. J., Blumenstock, J., Bjelland, J., Engø-Monsen, K., De Montjoye, Y.-A., & Iqbal, A. M. (2017). Mapping poverty using mobile phone and satellite data. *Journal of The Royal Society Interface*, 14(127), 20160690.
- Tan, M., & Le, Q. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. *International conference on machine learning*.

Yeh, C., Perez, A., Driscoll, A., Azzari, G., Tang, Z., Lobell, D., Ermon, S., & Burke, M. (2020). Using publicly available satellite imagery and deep learning to understand economic well-being in Africa. *Nature communications*, 11(1), 2583.